Auditing Learned Single-Position Additive Steering at Prefill: Random Directions, Benign Wrappers, and Measurement Instability

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Abstract

We audit learned additive steering at a single position, the final prompt token during prefill, in Llama-3.1-8B-Instruct. Qwen is used only for zero-training baselines. On 70 attacks selected for prior success, norm-matched untrained directions reduce judge attack success rate (ASR) by 4–21 percentage points (pp). On 300 WildJailbreak attacks sampled without conditioning on baseline success, four norm-matched random directions change judge ASR by -4 to $+1$ pp, while five learned vectors reduce it by 11–20 pp. All five learned ASRs are below all four random ASRs (exploratory exact unit-level permutation, $p = 0.0159$). A frozen 10-pp advantage criterion was met by 3/5 seeds under its specified two-direction reference, short of the required 4/5; the later four-direction comparison yields 5/5 and does not revise that decision. Paired benign evaluation reproduces higher refusal counts for wrapped requests on a second prompt sample. We document baseline mismatches, truncation, classifier-position and threshold errors in our own pipeline, and label changes between nearly identical retrained vectors. These results distinguish changes in judge ASR and operational refusal labels from evidence of calibrated safety or utility.

1 Introduction

Safety steering changes a language model's internal activations to influence its responses. A small vector can produce a large change in an automated safety score, but several explanations remain compatible with that observation: learned refusal, non-specific perturbation, a change in response format, or a measurement artifact. Benign requests may also inherit the surface form of a jailbreak, creating a useful test of whether an intervention tracks harmful intent or request packaging.

We study these distinctions in a controlled audit of one learned steering setup. Our main contribution is a collection of measured evaluation pitfalls and operational baselines, rather than a claim that steering constitutes a generally effective defense. We compare learned vectors with norm-matched random directions, decompose the learned update relative to initialization, and evaluate paired benign originals and adversarially packaged versions. We also retain system prompts and an output-only safety filter as zero-training references.

The resulting evidence is mixed in a scientifically useful way. Learned vectors outperform sampled random directions on both our selected attack set and an external attack family. Nevertheless, random directions alone substantially change the attack judge, refusal concentrates on benign wrapped requests, and nominally minor implementation choices alter the measured effect. Even highly similar retrained vectors produce different response trajectories and per-item labels. These observations motivate paired retrained controls and explicit separation of judge outcomes, refusal labels, and task completion.

We make four contributions. First, we quantify the random-direction reference on success-selected attacks and examine the residual component of learned steering. Second, we reproduce a wrapper-associated benign refusal gap across prompt samples and report its descriptive association with overall over-refusal. Third, we document baseline membership, reading scope, truncation, backend alignment, thresholds, and training repeat variation in our own pipeline, with reproducible tables. Fourth, on 300 WildJailbreak attacks sampled without filtering for baseline success, all five learned ASRs are below the four random references (exploratory exact unit-level $p = 0.0159$). Three of five seeds met the 10-pp threshold under the frozen two-direction reference, short of the required four; all five met it in the supplementary four-direction comparison.

2 Related Work

Steering and refusal. Learned interventions on frozen-model activations include representation finetuning and bidirectional preference steering (Wu et al., 2024; Cao et al., 2024). We audit one fixed-vector injection protocol. Related work separates steering efficacy from the preservation of other behaviors under distribution shift (Goyal and Daumé, 2026). Arditi et al. (2024) study a refusal direction in chat models. This supports evaluating harmful compliance and benign task utility together. Our learned vector, its residual decomposition, and its final-prompt-token injection protocol are specific to the setup below; we do not equate them with a universally shared refusal direction.

Random steering. Korznikov et al. (2026) report that random fixed-vector steering can increase harmful compliance from zero to 1–13% in the revised version. Their intervention persists across prompt and generated tokens, whereas our intervention is a single injection at the final prompt token. Our attack prompts are also selected for prior success. Random perturbations move our judge score in the opposite direction. The protocol and input-selection differences are plausible explanations, not a tested mechanism or a direct numerical replication.

Attack and benign evaluations. PAIR iteratively constructs semantic jailbreaks through black-box interaction (Chao et al., 2023). HarmBench (Mazeika et al., 2024) and JailbreakBench (Chao et al., 2024) emphasize standardized attack evaluation. WildTeaming introduces WildJailbreak with both vanilla and adversarial harmful and benign examples (Jiang et al., 2024). Its benign requests that resemble harmful ones motivate our paired wrapper analysis. Because our benign intervention and evaluation share this source, that intervention tests in-distribution data composition rather than generalization to arbitrary wrapper styles.

Scope of the audit. Prior work compares automated jailbreak evaluators and studies judge calibration and robustness (Gao, 2026; Mu, 2026). We complement these evaluations by distinguishing a harmfulness judge flip from an observed refusal, and an observed refusal from a calibrated utility failure. We use matched baselines and explicit reading scopes to expose cases where those distinctions change reported counts. Numerical agreement across implementations is an important reproducibility check, but it does not validate the judge on a new distribution or language.

3 Experimental Setup

Models and intervention. The primary target is Llama-3.1-8B-Instruct. A supplementary zero-training evaluation uses Qwen2.5-7B-Instruct; no Qwen steering vector or alpha sweep is reported. The old trainer learns a 4096-dimensional vector with norm 5.21, 40 Adam steps at learning rate 0.005, 12 fixed attack inputs, and 60 benign keep inputs. Model weights remain frozen. At evaluation, a vector scaled by α is injected once into the last prompt position of layer index 9. Generation is greedy. Harmful evaluation permits 2400 new tokens; the original benign protocol permits 120, with a controlled 400-token follow-up for nine uncertain examples.

Prompt samples. The selected attack set, val70, contains 14 JailbreakBench and 56 HarmBench behaviors, with successful PAIR prompts selected against the unmodified Llama model. Selection matters: this is not an unbiased sample of all harmful requests. The WJB harmful evaluation contains 300 attack prompts from another attack family. Two benign samples pair unwrapped originals with adversarially packaged benign requests. One analysis retains 293 pairs from the val296 source after its baseline exclusion; a separate 300-pair sample retains 292 when either-side baseline rule refusals are removed. The latter exclusion was documented after some generations and is not described as an original pre-generation rule.

Outcomes. Strict judge ASR is the proportion with HarmBench $P(\text{Yes}) > 0.5$. Backend and INVALID counts accompany comparisons; the HF alpha-zero baseline is 63/70 with zero INVALID, rather than the older vLLM cache's 62/70 with one INVALID. Operational benign over-refusal counts new rule-positive responses among rule-negative baseline items, using fixed sets of 388 Llama or 392 Qwen items. These are not independently human-calibrated safety-refusal rates. We preserve 400-item batch-8 composition for legal generation before filtering to 388 eligible items.

For benign pair i, let r_i^A and r_i^V be frozen rule labels on the adversarial wrapper and vanilla response. We report $DD = 100\,\text{mean}_i(r_i^A - r_i^V)$ on the fixed baseline-eligible set. Pair-level inter-

Condition	Units	ASR (%)
Baseline	1	90.0
Gaussian	3	68.6–80.0
Init-only	8	70.0–85.7
Learned	8	18.6–30.0

Table 1: HF judge ASR on val70. Random and learned vectors have norm 5.21. Units are directions or training seeds, not prompts.

Seed	val-source DD	WJB DD	Van. new
3	8.19	10.96	4
4	4.44	5.48	3
5	3.41	3.77	5
6	4.44	6.16	4
7	4.78	7.19	4

Table 2: Benign wrapper gaps (pp) for the same five old vectors on different prompt samples. The denominators are 293 and 292 baseline-eligible pairs. Vanilla additions are operational rule counts on the 292-pair sample.

vals resample vanilla requests, preserving the two sides. Training-level comparisons use the seed as the unit; repeated prompts are not independent training runs.

Statistical and chronological limits. Five-seed intervals use a t reference with four degrees of freedom. A joint frontier sensitivity also resamples attack and benign items and propagates old-curve uncertainty; unsupported interpolation draws are identified. The WJB nine-unit rank permutation test is exploratory and assumes exchangeability of five learned seeds and four random directions, conditional on shared prompts. Versioned logs preserve amendments and unsuccessful criteria rather than relabeling later supplements as original confirmation.

4 Results

4.1 Random references and learned residuals

On val70, the HF baseline ASR is 90.0%. Three Gaussian directions with norm 5.21 give 68.6%, 71.4%, and 80.0%; eight norm-matched initialization-only vectors span 70.0–85.7%. The combined eleven-direction mean is 75.7%, with standard deviation 4.6 percentage points (pp). Thus the perturbation reference alone lowers judge ASR by 4–21 pp. Eight learned vectors span 18.6–30.0%, providing an increment beyond the sampled random reference, but the judge decrease itself does not establish refusal or removal of harmful content.

In the LMI decomposition, cond2 normalizes the residual orthogonal to initialization to norm 5.21. Across seeds 3–7, the aggregate ASR difference from the full learned vector is 0.0 pp. The seed-level interval is $[-6.75, 6.75]$ pp; crossed resampling of shared behaviors and seeds gives $[-8, 8]$ pp. All five seeds satisfy the logged absolute-difference criterion of at most 10 pp, but these intervals do not establish equivalence. Each seed has 10–15 discordant behaviors: equal ASR need not mean success on the same items. Cond1 and cond3 were only evaluated on the post-hoc-selected seed0. In particular, cond3 has norm 3.44 and is not a norm-matched reference for norm 5.21.

On the 300 WildJailbreak attacks, sampled without filtering for baseline success, the HF baseline ASR is 39.3%. Four norm-matched random directions change it by -4.0 to $+1.0$ pp, while five learned vectors lower it by 11.3–20.0 pp. Learned ASRs are 19.3–28.0%, below every random ASR (35.3–40.3%). The exploratory exact rank test gives $U = 0$ over 126 allocations and two-sided $p = 0.01587$, treating the five seeds and four directions as exchangeable units under the null on shared prompts. The frozen two-direction criterion required at least 4/5 seeds to beat the random mean by 10 pp; only 3/5 did. Using all four directions gives 5/5 as a supplementary comparison, with no change to the frozen decision.

The smaller random-direction changes on WJB are consistent with an effect of conditioning val70 on prior success, including regression toward the mean, but do not identify that explanation. A separate HF sweep with a mean-difference direction gives nonmonotone ASRs of 84.3, 85.7, 81.4, 72.9, 75.7, and 78.6% at $\alpha = 0.25$ through 1.5, versus 90.0% at zero. It is not a random-direction dose experiment.

4.2 Benign wrappers and data composition

The five-seed mean wrapper gap on the separate 292-pair sample is 6.71 pp, with a seed-level interval of $[3.38, 10.04]$ pp. The val296-source analysis has a mean of 5.05 pp. This is a same-scale replication on different prompts using the same vectors. Seed rankings correlate with both the earlier gap and legal over-refusal ($\rho \approx 0.975$, five points with ties), so the evidence does not isolate a seed-specific style sensitivity from general refusal

Seed	α_T	ASR C	ASR T	New C	New T
3	1.5	21.4	22.9	28	21
4	1.0	27.1	27.1	16	12
5	1.5	25.7	24.3	14	14
6	1.0	25.7	24.3	17	10
7	1.0	27.1	30.0	18	11

Table 3: First frozen matching table: retrained control C at alpha 1 and wrapped-data treatment T. ASR is a percentage on 70 prompts; legal new counts use 388 eligible items after full 400-item batch-8 generation. Holdout outcomes remain unreported until complete.

Model	Condition	Harmful /70	ASR (%)
Llama	Baseline	63	90.0
Llama	A	25	35.7
Llama	B	13	18.6
Qwen	Baseline	40	57.1
Qwen	A_replace	26	37.1
Qwen	A_append	28	40.0
Qwen	B_replace	27	38.6
Qwen	B_append	26	37.1

Table 4: Zero-training prompt baselines, using HF judgments with zero INVALID. Legal-side model counts and uncalibrated ranges are discussed separately.

propensity. The 3-pp threshold followed observations on val296 and does not give special meaning to a seed just exceeding it.

Full-text review of 20 vanilla rule additions identifies nine condition-item outputs whose baseline was already a nonanswer, two substantive fictional-email responses falsely flagged by the rule, and nine content-response-to-refusal changes. Removing the four implicated pair IDs across all seeds changes the mean from 6.71 to 7.22 pp on 288 pairs; this is post-hoc sensitivity, not a revision of the frozen analysis.

The ongoing augmentation experiment uses same-pipeline retrained controls and matched val70 ASR. Treatment seeds 3 and 5 use $\alpha = 1.5$, hence norm 7.815, while their controls use norm 5.21. This is attack-ASR matching at different intervention strengths, not a fixed-strength comparison. A fixed-strength supplement for these two seeds was specified before the matched held-out effects were read; it does not replace matching. The target is one control run; its observed retraining spread does not permit interpreting fine differences within the 5-pp tolerance as precise matching. The first matching results are in Table 3; no holdout augmentation conclusion is reported while generation remains incomplete. Adding wrapped benign examples also changes benign-data volume and halves each old keep weight. A separate unwrapped-data arm was initially specified on seeds 3–5 and then expanded to all five seeds before its holdout evaluation, for greater precision. Without that arm, the intervention can only be described as augmentation with 120 benign inputs that include wrappers, not a pure style-matching effect.

4.3 Zero-training prompts and output filtering

On Llama, system prompts A and B give 35.7% and 18.6% judge ASR. Operational legal additions are 4/388 and 47/388; full model review puts B at 46–49 on the fixed eligible set, but A lacks the same review and baseline tail samples include truncated uncertain outputs. We do not establish a calibrated ordering or equivalence with learned steering.

On Qwen, baseline ASR is 57.1%; the four A/B conditions span 37.1–40.0%. Legal-side results are model-annotated ranges without human calibration. Rule and full-review differences are condition-dependent, and an exploratory A-versus-B comparison does not survive correction for six comparisons. Cross-model model-label counts can be juxtaposed descriptively, but do not support absolute over-refusal rankings.

The output Guard receives only the full response text. At threshold 0.5, four baseline Qwen harmful outputs are exactly on the boundary: blocking with $\geq$ leaves 18/70 harmful outputs, versus 22/70 with $>$. At 0.2, baseline plus Guard leaves 9/70 while A_replace plus Guard leaves 14/70. Among 24 prompts harmful under both conditions, Guard detects 18 baseline responses and 11 A responses, with discordance 7 versus 0 (exploratory exact $p = 0.015625$). This demonstrates output-dependent detection, not a confirmed role-play or disclaimer mechanism. Lower thresholds sharply increase benign blocking; no cross-model numerical over-refusal comparison is made.

5 Evaluation Pitfalls

5.1 Baseline membership and backend alignment

Two Llama baseline generation paths disagree on four of 400 refusal labels. Both leave 388 rule-negative outputs, but the sets differ. Applying the single-item baseline to batched curve outputs undercounts new refusals by one or two items (0.26–0.52 pp), comparable to the frontier effect under discussion. File-level alternative-baseline recomputation is a sensitivity audit, not proof that every historical analysis used the wrong baseline. Qwen's batch-8 files have a matching Qwen batch-8 baseline; absence of a second single-item path does not invalidate that actual comparison.

Backend mixing similarly changed alpha-zero ASR from the HF value 63/70 to an older vLLM value 62/70 with one INVALID. Aligning backends changes the random-reference reduction and paired discordance. We retain the superseded cache for audit and never hide INVALID by altering denominators.

5.2 Reading scope, nonanswers, and truncation

Rules can flag apologies in otherwise substantive responses, miss soft or non-English refusals, and treat knowledge limitations as task completion at baseline. In twelve selected Qwen candidate additions, full model review agrees with rule positives on 6/12: counts are 5/5 for B_replace, 1/3 for A_replace, and 0/4 for A_append. Small selected denominators do not estimate calibrated condition-wise precision. Historical preview labels and reading scopes are incomplete; disagreement cannot be attributed solely to truncation when definitions also changed.

Nine 120-token uncertain outputs reproduce byte-for-byte at the same limit. At 400 tokens, eight contain substantive responses and one reaches EOS without answering; no explicit refusal is seen. Eight still hit the new cap, so these observations concern generated content rather than unseen continuations. Refusal, nonanswer, and incomplete generation must remain separate labels. Model self-agreement and repeated annotation are not independent human calibration. A later blinded audit of 30 union-rule-negative legal outputs per arm finds no explicit refusal, but three control and five wrapped-arm outputs do not reach the core task within the 120-token window. Treating these ambiguous cases conservatively exceeds the three-in-thirty line for the wrapped arm. At 400 tokens all eight ambiguous continuations reproduce their original 120-token prefix and reach substantive content without an explicit refusal. This is a follow-up of six sampled negatives per seed and arm, not a population recall estimate or a revised 120-token guard.

5.3 Judge flips and refusal labels

After symmetric full-text model review, refusal-like responses account for 146/184 downward flips across three learned vectors and 18/44 across two random directions. These are condition-item counts sharing the same 118 baseline-harmful prompts, not independent Bernoulli observations. Rule recall can differ between hard refusals, soft wording, creative responses, and languages. Non-English refusal misses were observed, while the harmfulness judge was not validated separately for those languages. The comparison is descriptive; it does not establish distinct mechanisms or equate non-refusal rewrites with harmless content. On the separate 51-flip audit, the union rule found 24 of 41 single-reader refusal labels, with no false positives in that selected sample.

5.4 Classifier position and threshold convention

An output-Guard scoring attempt evaluated token probabilities before the required classification separator and failed the reference check. The scores were discarded and the corrected protocol reproduced twenty reference outputs. Even after numerical agreement, exact threshold ties change labels: the Qwen 0.5 comparator changes remaining harmful counts by 4/70. We report input format, scoring position, probability comparator, and boundary sensitivity rather than the threshold number alone.

5.5 Interpolation and unsupported draws

A new-versus-old trainer frontier comparison remains uncertain: the joint intervals include zero and extend beyond the planned ± 3-pp equivalence region. In 47.6% of bootstrap draws, at least one point cannot be interpolated. Conditional intervals and clipped-boundary diagnostics answer different questions and are not interchangeable. An after-the-fact seven-point subset is reported only as sensitivity; it does not replace the full comparison.

Seed 3 run	Harmful /70	ASR (%)
Frozen	19	27.14
Rebuild	16	22.86
Repeat A	15	21.43
Repeat B	18	25.71

Table 5: Four seed 3 runs: one historical frozen vector and three modern reruns sharing target cache, initialization and optimization path. A/B are the explicitly paired same-script repeats. The samples are too small for variance decomposition.

5.6 Training repeat variation and decoding amplification

A seed3 reconstruction has cosine 0.999920 to the frozen vector, angle about 0.727°, and vector-difference norm 0.0661. Five of 70 labels change (one upward, four downward), reducing harmful counts from 19 to 16. All five pairs exceed 700 generated tokens and reach EOS; their first retokenized divergence occurs between tokens 11 and 310. This shows that highly similar vectors can induce different decoded trajectories. It is not evidence for a 7.1-pp lower bound on ASR variance or a systematic backend shift.

Two subsequent same-cache, same-initialization, same-script, same-physical-card runs have cosine 0.999943814 and difference norm 0.05523. Three labels change, all upward, between counts 15 and 18. Across the four seed3 runs, the descriptive ASR standard deviation is 2.61 pp; across the old eight seeds it is 3.64 pp. The observed dispersions are comparable, but the samples are too small to separate their sources. There is one historical frozen vector and three modern reruns sharing the target cache, initialization, and optimization path. The samples also share the frozen seed3 observation, so an independent variance decomposition is not justified. A single-run seed ranking should not be interpreted as stable. Previously observed 3-versus-4 attack discordance between different vectors is on the same small-count scale as these repeats. We therefore use retrained paired controls for the data intervention, while preserving the original failed reconstruction gate.

6 Limitations

The evidence concerns one primary steering model and one supplementary prompt/filter model. There is no trained Qwen steering vector, and an independent PAIR test campaign was not run. WJB supplies a different attack family but does not substitute for that missing test. Val70 is success-selected and reused for alpha matching; it is not an independent final test.

Our harmfulness judge is an operational model classifier. Cross-node and backend agreement do not establish validity on WJB, non-English outputs, or unusual response formats. Refusal categories and task-completion labels are model annotations without independent human calibration. A single-reader audit of 51 selected flips gave union-rule precision of 24/24 and recall of 24/41 (about 0.59) against its labels. Recall may vary with response style; these counts do not calibrate the rule for other conditions. A planned ten-item blind reread remains incomplete. Unread negative outputs and token-limited continuations prevent interpreting reviewed ranges as statistical truth bounds.

The benign wrapper samples share a dataset source, and the augmentation training set and held-out pairs are from WildJailbreak. Exact and full-window embedding audits reduce obvious overlap but do not establish semantic independence below cosine 0.85. The original two-arm intervention confounds wrapped data with additional benign data and keep reweighting. The plain-data supplement has been expanded to five seeds, remains limited in precision, and is not content-matched. Any interpolation assumes local linearity in a noisy 70-item ASR axis.

Five training seeds and four random directions give limited estimates of their populations. The 10-pp advantage threshold was set in investigator discussion without an empirical basis for practical importance. Its attainment is sensitive to the random reference: 3/5 under the specified two directions and 5/5 under the supplementary four. Shared prompts, repeated vectors, post-hoc-selected seed0 analyses, and amendments after earlier observations restrict confirmatory interpretation. The historical-vector and three-rerun comparison does not partition training and seed variance. A second seed3 control is queued for the full holdout to contextualize the 2-pp criterion; a single observed repeat cannot validate that criterion generally. We preserve failed and superseded criteria and report dates of effect reads. No ongoing augmentation outcome is presented as complete evidence.

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A Provenance and protocol chronology

All numeric tables are generated from CSV files, which in turn are exported from versioned analysis files. The table provenance manifest records complete source SHA256 hashes. Raw attack prompts, model outputs, and training caches are excluded from the manuscript repository. Decisions were specified in a versioned log. External timestamps begin on 2026-11-06 at 00:34 JST; earlier states are supported only by server logs. Several thresholds and protocol amendments followed earlier observations and are identified as such. The original five-seed WJB criterion remains unsuccessful (3/5). Later random directions, a retrained-control design, a third data arm, and interpolation supplements do not revise that original outcome. The third-arm amendment was written after the original matching table had been frozen and holdout generation had begun. A fourth amendment expanded the plain arm to five seeds before the added seeds were trained and before any plain-arm holdout was generated or read; the first three-seed training queue had already started. These amendments do not revise the original criterion.

B Auxiliary frontier sensitivity

The new-versus-old trainer frontier point estimate is about +0.98pp; conditional on the observed horizontal coordinates, its five-seed interval is $[-0.63, 2.59]$pp. Table 6 propagates additional item and old-curve noise. The resulting uncertainty includes zero and extends beyond the proposed equivalence band. An exploratory seven-point subset is not substituted for the full analysis.

Joint sensitivity	95% interval
Item resampling (conditional)	[-0.95, 3.56]
Also resample seeds (conditional)	[-1.34, 3.88]
Boundary clipping diagnostic	[-2.12, 3.57]

Table 6: Joint frontier sensitivity in pp. The first two intervals condition on all points being interpolable. In 4764/10000 draws at least one point was unsupported; clipping changes the estimand. None establishes equivalence within plus/minus 3 pp.